

Problem 2.19

Show that, for any amount function $A(t)$, we have

$$A(n) - A(0) = \int_0^n A(t) \delta_t dt.$$

Solution

We know that

$$\delta_t = \frac{A'(t)}{A(t)}.$$

Substituting this into the integral, we obtain

$$\int_0^n A(t) \delta_t dt = \int_0^n A(t) \cdot \frac{A'(t)}{A(t)} dt.$$

Cancelling $A(t)$, we get

$$\int_0^n A(t) \delta_t dt = \int_0^n A'(t) dt.$$

Now integrate:

$$\int_0^n A'(t) dt = A(t) \Big|_0^n = A(n) - A(0).$$

Therefore,

$$\boxed{A(n) - A(0) = \int_0^n A(t) \delta_t dt.}$$